JJJ-B-81E
October 19, 1976
SUPERSEDING
Int. Fed. Spec. JJJ-B-0081D(GSA-FSS)
March 26, 1965 and
Fed. Spec. JJJ-B-81B
February 27, 1959

FEDERAL SPECIFICATION

BALL, ABSORBENT, COTTON OR RAYON, NON STERILE

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers absorbent balls made of cotton or viscose rayon fibers, for use as medical wipes or sponges.

1.2 Classification. The absorbent balls covered by this specification shall be of the following sizes, as specified (see 6.2):

Size 1 - Medium
Size 2 - Large

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form part of this specification to the extent specified herein:

Federal Specifications:

PPP-B-636 - Boxes, Shipping, Fiberboard.

Federal Standards

Fed. Std. No. 102 - Preservation, Packaging, and Packing Levels.
Fed. Std. No. 123 - Marking for Shipment (Civil Agencies).
Fed. Test Method Std. No. 191 - Textile Test Methods

FSC 6510

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(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, US Government Printing Office, Washington, DC 20402.)

(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, DC, Atlanta, Chicago, Kansas City, MO, Forth Worth, Denver, San Francisco, Los Angeles, and Seattle, WA.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Standards:

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	-	Marking for Shipment and Storage

(Copies of Military Specifications and Standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposa shall apply.

The United States Pharmacopeial Convention, Inc. (U.S.P.)

Pharmacopeia of the United States

(Application for copies should be addressed to the Mack Publishing Company, Easton, PA 18042.)

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3. REQUIREMENTS

3.1 Material. The basic material may be either cotton or rayon fibers complying with the requirements of table I and as specified herein. The finished balls shall be so constructed that there shall be little or no separation or loss of fibers when tested in accordance with 4.4.9. Fiber length shall be tested as specified in 4.4.1. Shall be free of lumps (including lumps consisting of compacted fibers), oil spots, streaks, dirt and foreign material.

TABLE I. Material requirements

Characteristic	REQUIREMENT		Test
	Cotton	Rayon	
Ash	0.2 percent maximum (max.)	2.0 percent max.	4.4.2
Alkalinity or acidity	No pink color		4.4.3
Dyes	No blue or green color		4.4.4
Fatty matter	0.7 percent residue max.		4.4.5
Water soluble substances	0.35 percent max.	1.0 percent max.	4.4.6
Absorbency	Submerge in 10 seconds and shall not retain less than 20 times its weight of water		4.4.7

3.1.1 Cotton. Cotton shall be purified, new or new reworked cotton, such as combed or carded strips; white, fully bleached and shall contain no cotton linters or cotton which has been reworked from yarn, threads, or fabric. Not less than 40 percent by weight of the fibers of the cotton shall be 12.5 millimeters (mm) or greater in length, and not more than 20 percent by weight, of the fibers shall be 6.25 mm or less in length.

3.1.2 Rayon. Rayon shall be regenerated viscose rayon fiber, fully bleached, clean, dull and crimped. Not less than 70 percent by weight of the rayon fibers shall be 19 mm or greater in length, and not more than 5 percent by weight of the fibers shall be 6.25 mm or less in length. Filaments shall be not greater than 3.0 denier.

3.2 Fabrication. The absorbent ball shall be approximately spherical or barrel shaped, shall be machine made and uniform in size.

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3.3 Weight. The average weight of the absorbent balls, when determined as specified in 4.4.10, shall comply with the requirements in table II.

TABLE II. Dimensions and weight

Size	Nominal diameter (inches)	Actual diameter (inches)	Grains per ball (average)	
			Minimum	Maximum
Medium	1-1/4	1-1/8 ± 1/8	3-1/2	5-1/2
Large	2	1-5/8 ± 1/8	12-1/2	17-1/2

3.4 Moisture content. The absorbent balls shall have a moisture content of 5.5 to 12 percent for cotton and 6 to 12 percent for rayon when tested in accordance with 4.4.8 before packaging.

3.5 Workmanship. The workmanship shall be first class throughout. The absorbent balls shall be free from defects which detract from their appearance or affect their serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of three years after delivery of the supplies to which such records relate.

4.1.2 Inspection Inspection, as used in this specification, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

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4.1.3 Certificate of compliance. Contractor shall submit a certificate of compliance, stating that the manufacturing areas are humidified and the absorbent balls when tested before packaging, have a moisture content of 5.5 to 12% for cotton, and 6 to 12% for rayon.

4.1.4 Certificate of quality. Certificate of quality, supplied by the manufacturer of the cotton or rayon material, may be furnished in lieu of actual performance of such testing by the contractor, provided lot identity has been maintained and can be demonstrated to the Government. The certificate shall include the name of the contractor, the contract number, the name of the manufacturer or supplier, the NSN, the Item Identification, the name of the component/material, the lot number, the lot size, the sample size, the date of testing, the test method, individual test results, and the specification requirements.

4.2 Sampling.

4.2.1 For examination. Sampling for examination shall be conducted in accordance with MIL-STD-105, and table III. The unit of product for sampling purposes shall be one absorbent ball.

TABLE III. Sampling for examination

	Inspection level	AQL (percent defective)
For visual examination		
Major defects	II	1.0
Total defects (Major and Minor combined)	II	4.0

4.2.2 For tests. Sampling for tests shall be conducted in accordance with MIL-STD-105 and table IV, and as specified herein. Each test conducted in accordance with the U.S.P., shall be run twice per lot of absorbent balls and the lot shall be rejected when one individual test fails. The weight test shall be conducted 12 times per lot of absorbent balls and the lot shall be rejected when two or more individual weighing tests fail. No more than 200 absorbent balls are to be selected from a unit package. Each of the remaining tests shall be run six times per lot of absorbent balls and the lot shall be rejected when an individual test fails. Sampling shall be done in a random manner.

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TABLE IV. Sampling for tests

Component	Characteristic	Requirement	Test
Material	Fiber length	3.1	4.4.1
	Ash	"	4.4.2
	Alkalinity Acidity	"	4.4.3
	Dyes	"	4.4.4
	Fatty matter	"	4.4.5
	Water soluble substances	"	4.4.6
	Absorbency	"	4.4.7
Absorbent balls	Moisture	3.4	4.4.8
	Non-separation	3.1	4.4.9
	Weight	3.3	4.4.10

4.3 Examination. The absorbent balls shall be examined to determine compliance with all requirements contained within this specification.

4.3.1 Classification of defects. Examination shall be in accordance with table V. Examination shall not be restricted to the classified possible defects listed in table V.

TABLE V. Classification of defects

Categories	Defects
Major	
101	Balls not clean and free from foreign matter.
Minor	
201	Balls in the lot not uniform in size and shape.
202	Balls not within the size limits specified.

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4.4 Tests.

4.4.1 Fiber length. Fiber length shall be determined as specified in the U.S.P. for fiber length of cotton.

4.4.2 Ash. Ash shall be determined as specified in the U.S.P. for residue on ignition of purified cotton.

4.4.3 Alkalinity-acidity. Alkali and acids shall be determined as specified in the U.S.P. for alkalinity of purified cotton.

4.4.4 Dyes. Dyes shall be determined as specified in the U.S.P. for dyes of purified cotton.

4.4.5 Fatty matter. Fatty matter shall be determined as specified in the U.S.P. for fatty matter in purified cotton.

4.4.6 Water-soluble substances. Water soluble substances shall be determined as specified in the U.S.P. for water-soluble substances of purified cotton.

4.4.7 Absorbency. Absorbency shall be determined as specified in the U.S.P. for absorbency of purified cotton.

4.4.8 Moisture. The absorbent balls shall be tested for moisture in accordance with method 2600 or method 2601 of Fed. Test Method Std. No. 191.

4.4.9 Test for non-separation. Two balls shall be tested, one dry and one wet with water, with the excess water squeezed out. The ball shall be rubbed gently several times on the smooth, dry part of the forearm, and shall pass the test requirement in 3.1.

4.4.10 Weight. Average weight per ball shall be determined by weighing 100 balls and dividing the result by 100. Weight of balls shall be based on dry weight.

4.5 Examination of preparation for delivery. An examination shall be made to determine compliance with the packaging, packing and marking requirements of section 5. Defects shall be scored as specified in table VI. Sampling shall be in accordance with MIL-STD-105. The sample unit shall be one container fully prepared for delivery. The lot shall be the number of containers offered for inspection at one time. The inspection level shall be S-2 with AQL of 4.0 expressed in terms of percent defective.

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TABLE VI. Examination of preparation for delivery

Examine	Defect
Contents	Not as specified.
Container	Not as specified.
Marking	Omitted; incorrect; illegible; improper size, location, sequence or method of application.
Material	Component missing or damage.
Workmanship	Bulging or distortion of containers

4.5.1 Examination of closure, waterproofing and banding of container. When the shipping container is required to comply with PPP-B-636, the examination for defects in closure, water-proofing or banding shall be in accordance with the appendix of the box specification.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A or C as specified (see 6.2)

5.1.1 Level A.

5.1.1.1 Unit package.

5.1.1.1.1 Medium size. Four thousand (4000) shall be packaged in a heat sealed 2 mil polyethylene bag. The balls shall be compressed within the bag in a manner that will not impair utility of the product for the intended end use. The polyethylene bag shall be further packaged into a box conforming to PPP-B-636, class weather-resistant. Box shall be closed in accordance with the appendix of the box specification. Inside dimensions of the box shall be approximately 13-1/2 inches by 12 inches by 14 inches.

5.1.1.1.2 Large size. Two thousand (2000) shall be packaged as specified in 5.1.1.1.1 except the inside dimensions of the box shall be approximately 18 inches by 12 inches by 18 inches.

5.1.2 Level C. Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose.

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5.2 Packing. Packing shall be level A, B or C as specified (see 6.2).

5.2.1 Level A. Four (4) units of medium size balls or two (2) units of large size balls shall be packed in an exterior container designed for a type 2 load and constructed in accordance with PPP-B-636, class weather-resistant, grade V3c. Closure shall be method V as specified in PPP-B-636. Strapping shall be in accordance with the appendix of PPP-B-636.

5.2.2 Level B. Four (4) units of medium size balls or two (2) units of large size balls shall be packed in an exterior container designed for a type 2 load and constructed in accordance with PPP-B-636, type CF, class domestic. Closure shall be as specified in the appendix of PPP-B-636.

5.2.3 Level C. The subject commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.3 Marking.

5.3.1 Civil agencies. Marking for shipment shall be in accordance with Fed. Std. No. 123.

5.3.2 Military activities.

5.3.2.1 Unit package. Each unit package shall be marked in accordance with MIL-STD-129. Item identification shall include "cotton or viscose rayon," as applicable. In addition, marking shall include name and address of manufacturer, contract or purchase order number, date of manufacture in lieu of date packed and the legend: "CAUTION-NOT STERILE."

5.3.2.2 Exterior container. Exterior container shall be marked in accordance with MIL-STD-129 except that the date of manufacture shall be shown in lieu of the date packed.

6. NOTES

6.1 Intended use. The absorbent balls covered by this specification are intended to be used as medical wipes or sponges.

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Size required (see 1.2).
(c) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

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6.3 This specification is intended to cover the following items appearing in the Federal Supply Catalog:

National Stock Number	Item Identiciation
6510-00-201-4100	BALL, ABSORBENT, Cotton or Rayon 1-1/4 inch Diameter, 4000s
6510-00-201-4150	BALL, ABSORBENT, Cotton or Rayon 2 inch Diameter, 2000s

MILITARY CUSTODIANS:
Army - MD
Navy - MS
Air Force - 03

Preparing activity:
DSA-DM

Civil Agency Coordinating Activities
GSA-FSS
VA-DMS
HEW-HSM

Project No. 6510-0614

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DOCUMENT IDENTIFIER AND TITLE: JJJ-B-81E BALL, ABSORBENT, COTTON OR RAYON, NON STERILE

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

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A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

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